

Review of the sensors and health-based parameters required for Stress Management of Students

→ Analysis of the Biomedical sensors:

Sensors	Accuracy	Resolution	Sensitivity	Sampling rate	Response time	Noise	Stability	Operating range
MAX30100 (Heart rate sensor)	±2% SpO ₂ , ±3% Heart rate	16-bit	High (uses IR/Red LED)	Up to 100 Hz	Approx. 0.5 sec	Moderate	Needs Calibration	1.8 V to 3.3 V and upto 60°C
MCP9808 (High-accuracy digital temperature sensor)	±0.25°C (typical)	0.0625°C	High	Can be changed	Approx. 0.25 s	Low	Excellent	-40°C to +125°C
Grove GSR Sensor (Skin response)	Varies per skin type	10 bit (Analog)	Medium	From 10 to 50 Hz	Approx. 1 s	High	Changes with time	Depends on skin condition
MPU6050 (Balance and motion)	±2%	16 bit	High	100 to 1000 Hz	1 to 5 ms	Low	Depends on temperature	2.3-3.4 V, -40 to 85°C
APDS9960 (Gesture, Proximity and Light)	Almost 90%	8,16 and 20 bits	Medium	Close to 120 Hz	<100 ms	Moderate	Very stable	2.4-3.6 V, -40 to 85°C
MAX4466 (Microphone with amplifier)	Varies on setup condition	10-12 bit (Analog)	Very high (as gain changes)	-	<1 ms	Low (at high gain)	Drifts with change in gain	2.4-5 V, 20 Hz to 20 kHz range
AD8232 (ECG and heart activity)	±5% Heart rate error	10 bit ADC input	High (µV to mV range)	250-500 Hz	<100 ms	Moderate	Affected by motion	3.3-6 V, temperature varies by skin contact

→ Sensors eligibility for building a stress management-based medical device:

According to ISO/IEC standard, these are the core requirements to be checked, such as:

- ❖ ISO 13485:2016 (Quality management system (QMS) for medical device)
 - Ensures consistent design, development, and production of safe and effective medical devices.
 - Requires documentation, traceability, risk management, and corrective actions.
 - Harmonized with regulatory requirements like FDA 21 CFR 820 and EU MDR.
- ❖ ISO 14971:2019 (Risk management across device lifecycle)
 - Provides a structured risk analysis and mitigation framework across the device lifecycle.
 - Requires identification of hazards, risk estimation, evaluation, and control measures.
 - Includes managing residual risks and benefit-risk analysis, particularly for patient safety.
- ❖ IEC 62304:2006 (Software lifecycle for medical device software)
 - Outlines software development lifecycle, including planning, development, verification, and maintenance.
 - Classifies software into risk-based safety classes (A, B, C).
 - Ensures traceability and documentation of software safety and performance.
- ❖ IEC 60601-1:2005 + A1:2012 (Medical Electrical Equipment Safety)
 - Specifies general safety and essential performance requirements for electrical medical devices.
 - Covers electrical shock protection, insulation, mechanical stability, and temperature limits.
 - Mandates safe operation under normal and single-fault conditions.
- ❖ ISO 80601-2-61:2017 (Safety/performance for pulse oximeters)
 - Specifies accuracy and safety standards for SpO₂ and pulse rate monitoring devices.
 - Includes requirements for calibration, performance under motion, and low perfusion.
- ❖ ISO 80601-2-56:2017 (Clinical thermometers (applies to temperature sensors))
 - Sets requirements for temperature sensors used on humans, such as MCP9808.
 - Ensures accurate readings in clinical and home environments.
 - Defines response time, measurement range, and environmental conditions.
- ❖ IEC 62366-1: 2015 (Using medical devices more smooth and precisely)
 - Ensures devices are safe and effective to use, minimizing user error.
 - Requires user interface design, validation, and usability testing.
 - Critical for sensors involving gesture, voice, or manual placement.
- ❖ ISO 10993-1 (safe and biocompatibility)
 - Evaluates biological risks of materials that contact the human body (e.g., skin, mucosa, blood).
 - Requires toxicology, irritation, sensitization, and systemic effects testing.
 - Essential for surface contact sensors like ECG, GSR, and microphones.

*FDA-Food and Drug Administration.

*MDR-Medical Device Regulation.

*CFR-Code of Federal Regulations.

Source: [ISO Standards for Medical Devices: Ultimate List & Overview](#)

[Medical Device Design and Development as per ISO 13485:2016 - Pharmadocx Consultants](#)

Sensors	Parameters measured	Applicable ISO / IEC Standards	Applicable scale for Medical Device	Explanation
MAX30100	Heart Rate, SpO ₂ .	☐ ISO 13485:2016 ☒ ISO 14971:2019 ☒ IEC 62304:2006 ☒ IEC 60601-1:2005 + A1:2012 ☐ ISO 80601-2-61:2017 ☐ ISO 80601-2-56:2017 ☒ IEC 62366-1: 2015 ☐ ISO 10993-1	Partially Fits	Measures SpO₂ → requires ISO 80601-2-61, which most modules lack unless medically certified. Touches skin → needs ISO 10993, not always met in hobbyist-grade boards. Quality certification (ISO 13485) often absent for module-level parts.
MCP9808	Skin/Body Temperature	☐ ISO 13485:2016 ☒ ISO 14971:2019 ☒ IEC 62304:2006 ☒ IEC 60601-1:2005 + A1:2012 ☐ ISO 80601-2-61:2017 ☐ ISO 80601-2-56:2017 ☒ IEC 62366-1: 2015 ☐ ISO 10993-1	Partially Fits	If used with skin contact (e.g., wearable), ISO 10993 is needed and often missing. To qualify as a body temperature sensor, it needs ISO 80601-2-56, usually unmet. Common modules are not ISO 13485 certified.
Grove GSR	Galvanic Skin Response (Skin Conductance)	☐ ISO 13485:2016 ☐ ISO 14971:2019 ☐ IEC 62304:2006 ☐ IEC 60601-1:2005 + A1:2012 ☐ ISO 80601-2-61:2017 ☐ ISO 80601-2-56:2017 ☐ IEC 62366-1: 2015 ☐ ISO 10993-1	Does not fit	No biocompatibility (ISO 10993) despite skin contact. Not certified under ISO 13485 or 60601 family. No clinical-grade validation of readings or risks → fails ISO 14971. No safety evaluation or software quality control for medical use.

MPU6050	Motion, Orientation (Accelerometer, Gyroscope)	☒ ISO 13485:2016 ☒ ISO 14971:2019 ☒ IEC 62304:2006 ☒ IEC 60601-1:2005 + A1:2012 ☐ ISO 80601-2-61:2017 ☐ ISO 80601-2-56:2017 ☒ IEC 62366-1: 2015 ☐ ISO 10993-1	Fits	Doesn't touch the body → ISO 10993 not mandatory. Often used in medical-grade wearables (fall detection, posture correction) → certified under ISO 13485 & 60601-1 by many vendors. Embedded software control is manageable under IEC 62304. Not a diagnostic sensor → 80601 substandards not applicable
APDS9960	Proximity, Gesture, Ambient Light, RGB	☐ ISO 13485:2016 ☒ ISO 14971:2019 ☒ IEC 62304:2006 ☒ IEC 60601-1:2005 + A1:2012 ☐ ISO 80601-2-61:2017 ☐ ISO 80601-2-56:2017 ☒ IEC 62366-1: 2015 ☐ ISO 10993-1	Partially Fits	No direct body contact → ISO 10993 not critical. May qualify under 60601 if used near-patient → often passes EMI compliance. Not certified under ISO 13485 typically. May lack usability certification in gesture-based health devices.
MAX4466	Microphone (Voice, Sound Input)	☐ ISO 13485:2016 ☐ ISO 14971:2019 ☐ IEC 62304:2006 ☐ IEC 60601-1:2005 + A1:2012 ☐ ISO 80601-2-61:2017 ☐ ISO 80601-2-56:2017 ☐ IEC 62366-1: 2015 ☐ ISO 10993-1	Does not fit	General-purpose audio mic → not certified for medical use. No quality, safety, or usability documentation. No biological or electrical certification. Can be used <i>with</i> certified systems, but alone is not compliant.
AD8232 Gravity ECG	ECG, Heart Rate Variability (HRV)	☐ ISO 13485:2016 ☒ ISO 14971:2019 ☒ IEC 62304:2006 ☒ IEC 60601-1:2005 + A1:2012 ☐ ISO 80601-2-61:2017 ☐ ISO 80601-2-56:2017 ☒ IEC 62366-1: 2015 ☐ ISO 10993-1	Partially Fits	Clinical ECG standards like ISO 80601-2-47 apply but are not fulfilled. No body contact safety validation (ISO 10993). Lacks module-level ISO 13485 compliance. Often meets 60601, 62304 when

				built into devices.
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References:

- ❖ <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8432146/>
- ❖ ISO Official Page: <https://www.iso.org/standard/67963.html>
- ❖ Application in Devices: MAX30100 Design Guide (Maxim)
- ❖ MCP9808 Datasheet (Microchip): <https://ww1.microchip.com/downloads/en/DeviceDoc/25095A.pdf>
- ❖ MPU6050 Datasheet: <https://invensense.tdk.com/wp-content/uploads/2015/02/MPU-6000-Datasheet1.pdf>
- ❖ AD8232 Datasheet (Analog Devices):
<https://www.analog.com/media/en/technical-documentation/data-sheets/AD8232.pdf>
- ❖ GSR in Biomedical Devices: <https://www.mdpi.com/1424-8220/20/9/2630>
- ❖ APDS9960 Datasheet: <https://www.sparkfun.com/datasheets/Sensors/Proximity/APDS9960.pdf>
- ❖ MAX4466 Datasheet: <https://cdn.sparkfun.com/datasheets/BreakoutBoards/MAX4466.pdf>

→ Some collected research papers and their reviews:

1) Wearables for Stress Management: A Scoping Review

Article link: [\(PDF\) Wearables for Stress Management: A Scoping Review](#)

Wearable Devices:

-9 Smart Devices: Head band, scarf, Arm band, Smartband, Patch, Finger sensors, Smart Pin, Smart Chest Band, Smart necklace, Smart glasses.

-Commercial Devices: Empatica E4, Spire Stone, Stila and Apple Watch, and Doppel.

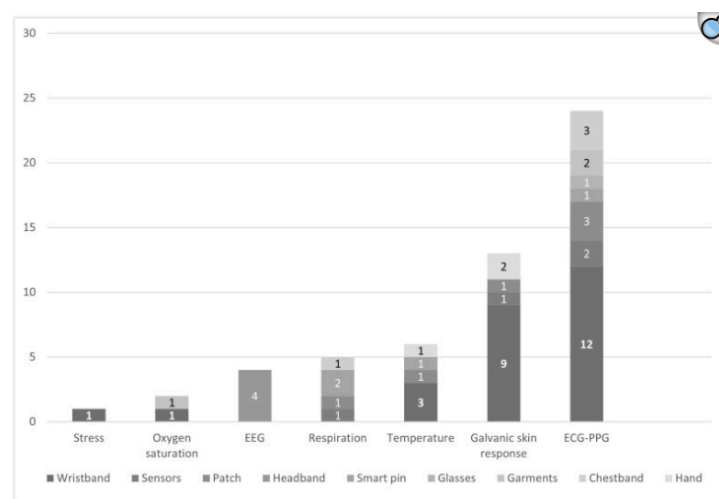
-Custom-made software and commercial software (Breathe, Life app, Stila, Wahoo)

-ECG and respiration sensors — HRV Biofeedback System:

Function: stress reduction and autonomic nervous system modulation.

Method: resonance frequency respiration training.

Stress Monitoring:



-Opportunistic and participatory sensing paradigms

Opportunistic Sensing:

- Data collected from specific stress symptoms
- Out of 40 studies conducted, 24 of them had Heart Rate (HR) and Heart Rate Variability (HRV) measured by ECG/PPG sensors.
- Respiration rate, EEG-based brain activity, galvanic skin response, temperature, oxygen saturation, stress.
- Other data such as User's walking behaviours, arousal, outburst patterns, sleep and posture.

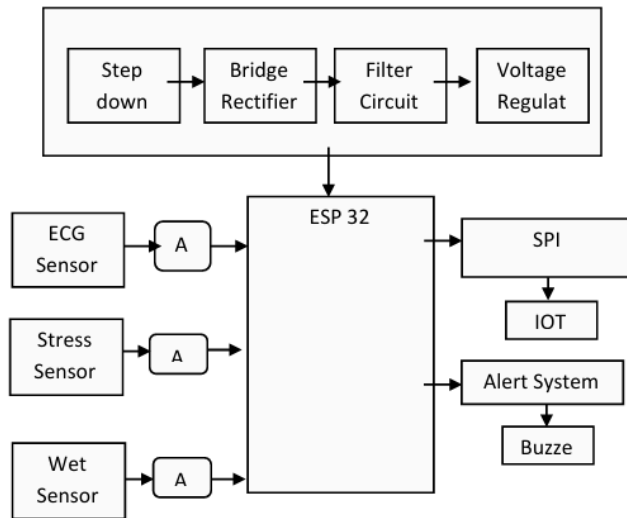
Participatory Sensing:

- Through apps and questionnaires inside them, users provide feedback on stress self-perception.
- Afterward, the system launches a questionnaire that users should complete to determine their stress self-perception.
- For example, Skulimowski and Badurowicz use horticulture therapy, which consists of taking care of a bonsai in a virtual reality game environment. Users must water, dust the leaves, and prune the virtual bonsai daily. At the same time, some physiological signals are measured to detect if users are stressed, which impacts the health of their bonsai since it starts to grow slowly and wither.
- Galvanic skin response is collected using a DTI-2 wristband and stored in digital calendars, like LifelogExplorer, to provide comprehensible interactive visualizations of users' arousal information in the context of their weekly life events.
- However, emerging studies share a focus on healthcare, particularly, regarding the relationship between Heart Rate (HR) and stress, which is the main interest of our study.

2) Stress detection and analysis from wearable device

Article link: [IJIRT164298_PAPER.pdf](#)

- Sensor selection, data collection, preprocessing, feature extraction, model development, validation, integration, user studies, and ethical considerations.
- Heart rate monitors, electrodermal activity sensors, and accelerometers are commonly used to capture physiological and behavioral indicators of stress.
- These features may include statistical measures, frequency domain features, time-domain features, and spectral features, among others.
- Machine learning or statistical models are developed for stress detection and analysis such as support vector machines, decision trees, random forests, and deep learning models.
- Performance metrics such as accuracy, sensitivity, specificity, and area under the curve are used to evaluate the models' performance.
- One of the disadvantages is that the study did not take into account other factors that could influence stress responses, such as psychological or environmental factors, and instead concentrated solely on physiological stress measures. Also the findings found were based on small sample size.



3) Pain and Stress Detection Using Wearable Sensors and Devices—A Review

Article link:

https://mdpi-res.com/d_attachment/sensors/sensors-21-01030/article_deploy/sensors-21-01030.pdf

-Parameters involved are pain and stress assessment, stress induction tests (such as Color-word Inference test, Hot water Immersion test, International Affective Picture system test etc), activities of heart, brain, muscle, electrodermal, pulse, skin etc. Behavioural signals such as speech, facial expressions, body movements, mobile phone usage etc were done.

-For pain detection, medical devices used here are Analgesia Nociception Index and Surgical Pleth Index.

Analgesia Nociception Index:

*Provides a quantitative measure (0–100) of parasympathetic nervous system tone.

*Based on electrocardiographic (ECG) data, particularly heart rate variability (HRV).

*Reflects the balance between sympathetic and parasympathetic activity.

*Useful for pain and stress assessment during and after surgery.

*Strong correlation with self-reported pain levels in postoperative periods.

*Especially suitable for non-communicative patients, e.g., under general anesthesia or critically ill.

*Applied in various conditions:

- ❖ Labor pain
- ❖ Laparoscopic abdominal surgery
- ❖ Post-volatile and opioid-based anesthesia

*Also being studied for its potential to reflect emotional states, not just physical pain.

*ANI can support objective, physiological pain monitoring, unlike subjective scales.

Surgical Pleth Index:

- *Measures sympathetic nervous system activity in response to painful surgical stimuli.
- *Generates a numerical index (0–100) combining:

- Photoplethysmographic (PPG) amplitude
- PPG pulse interval

- *Used during general anesthesia to assess response to analgesic medications.
- *Correlates with patient-reported pain scores (e.g., moderate to severe pain).
- *Still lacks standardized target value ranges for reliable decision-making.
- *Actively researched for its ability to predict postoperative pain levels.
- *Offers a real-time, non-invasive stress/pain monitoring tool in operating rooms.

- Wearable sensors used here are:

Type of signal	Commercialized Wearable Sensors used in Relevant Research	Wearable Sensors not yet Commercialized but used in Relevant Research
Heart activity	Empatica E4 wrist band, AutoSense, Cardiosport TP3, Wahoo chest belt, BioHarness 3	
Brain activity	MindWave mobile EEG headset	Device 1, Device 2
Muscle activity	DataLOG	Device 3
Electrodermal activity	Empatica E4 wrist band, BN-PPGED, Q-sensor, Shimmer sensor	
Respiratory	AutoSense, SleepSense	
Blood volume pulse/pulse plethysmograph	Empatica E4 wrist band, BN-PPGED	
Body/skin temperature	Empatica E4 wrist band, AutoSense	
Three-axis accelerometer data	Empatica E4 wrist band, Q-sensor	

